



Origin of magnetic moment and Angular momentum:

Let us consider an electron rotating in a circular orbit around the nucleus. The angular momentum of the electron in its orbit

$$\frac{h}{2\pi} = I\omega = mvr$$

$$\frac{h}{2\pi} = mvr \quad \text{--- (1)}$$

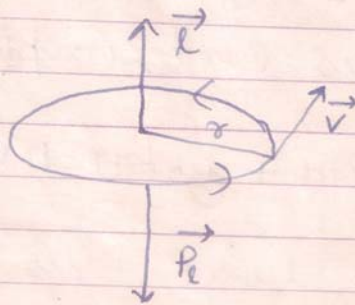


Where m = mass of electron.

v = Speed of electron in its orbit

r = radius of the circular orbit.

The direction of the angular momentum vector is given as follows:



"Curl the fingers of right hand in the direction of rotation the thumb gives the direction of the angular momentum vector $\vec{L}$."

An electron rotating in an orbit is equivalent to a current loop, the direction of current is opposite to the direction of rotation of the electron & the current loop gives rise to a magnetic field.

The magnetic moment of the current loop is given by ; $\mu = AI$



Where A = area of the loop.

I = current in the loop.

The current in the loop, due to the rotation of electron is given by $I = e \cdot \text{no. of rotations per sec.}$

$$I = e \cdot n = \frac{e\omega}{2\pi}$$

$$I = \frac{e}{2\pi} \frac{v}{r} \quad A = \pi r^2$$

Magnetic moment due to this orbital motion is

$$P_L = \pi r^2 \cdot \frac{ev}{2\pi r} = \frac{evr}{2} \quad (2)$$

This magnetic moment corresponding to the orbital motion having quantum no. L is a vector quantity, having direction given as follows
 "Curl the fingers of right hand in the direction of current (opposite to that of rotation of the electron) the thumb gives the direction of the magnetic moment vector i.e. $\vec{L}$ & $\vec{P}_L$ are oppositely directed.

Eliminating ' vr ' from eqⁿ (1) & (2)

$$\vec{L} = L \cdot \frac{h}{2\pi} = mvr \quad \therefore vr = \frac{\vec{L}}{m}$$

$$\therefore \vec{P}_L = \frac{e}{2} \frac{\vec{L}}{m} = \frac{e}{2m} \vec{L} \quad (3)$$

from (3): $\frac{\vec{P}_L}{\vec{L}} = \frac{e}{2m}$ this ratio is known as



Gyromagnetic ratio.

But $\vec{L} = \frac{lh}{2\pi}$ where l = Orbital quantum no. or angular quantum no.

$$|\vec{P}_L| = P_L = \frac{e}{2m} \frac{lh}{2\pi} = \left(\frac{eh}{4\pi m} \right) l \quad \text{--- (4)}$$

The quantity $\frac{eh}{4\pi m} = \mu_B$ = Bohr magneton

which is used as the unit of magnetic moment, in atomic or nuclear physics.

The value of Bohr magneton is

$$\mu_B = \frac{eh}{4\pi m} = \frac{1.6 \times 10^{-19} \text{ coulomb} \times 6.625 \times 10^{-34} \text{ J-sec}}{4 \times 3.142 \times 9.1 \times 10^{-31} \text{ kg}}$$

$$\therefore \boxed{\mu_B = 9.27 \times 10^{-24} \text{ amp. m}^2}$$

$$\therefore \boxed{P_L = \mu_B \cdot l} \quad \text{--- (5)}$$

Spin angular momentum and Spin magnetic moment:

We know that electron also possesses a spinning motion, having a spinning qu. no $\frac{1}{2}$

$\therefore$ The magnitude of the Spin angular moment : $|\vec{S}| = \frac{1}{2} \frac{h}{2\pi}$

The magnetic moment vector; due to the spinning motion of the electron will be opposite



in direction to that of the angular momentum vector.



However the ratio of the magnetic moment and the angular momentum of an electron due to spin is twice as large as that for orbital motion.

$$\text{if } \frac{\vec{P}_s}{\vec{S}} = 2 \times \frac{e}{2m} = \frac{e}{m}$$

$$|\vec{P}_s| = \frac{e}{m} |\vec{S}| = \frac{e}{m} \cdot \frac{1}{2} \frac{h}{2\pi} \quad (6)$$

Since an atom consists of several electrons and each has angular momentum and the mag. moment for both the orbital and the spinning motion, we can calculate total angular momentum and the magnetic moment of the atom. The direction of the magnetic moment is always opposite to the direction of angular momentum. Since the total contribution is a mixture of contribution from the orbital and spinning motion, the ratio of mag. moment and the angular momentum of the atom lies between

$$\frac{e}{m} \text{ to } \frac{e}{2m}$$

Therefore for the atom as a whole we can write $\vec{P} = g \cdot \frac{e}{2m} \vec{L}$



where 'g' is known as Lande 'g' factor and is the characteristic of the state of the atom.

In the nucleus of an atom the protons also move around in orbits & also have intrinsic spins and hence possess magnetic moment.

The magnetic moment of the nucleus is parallel to the angular momentum because protons being +vely charged particles direction of current is same as that of rotation.

For nucleus it is usual to write

$$\vec{P} = g \left(\frac{Ze}{2m_p} \right) \vec{L}$$

where m_p = mass of proton.

g = Nuclear 'g' factor.